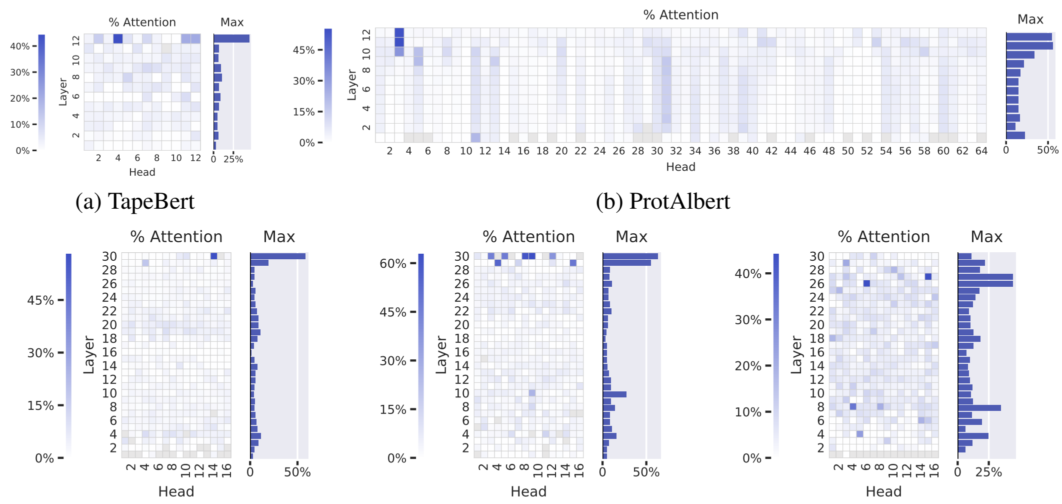


(a) TapeBert (b) ProtAlbert

(c) ProtBert (d) ProtBert-BFD (e) ProtXLNet

Figure 2: Agreement between attention and contact maps across five pretrained Transformer models from TAPE (a) and ProtTrans (b–e). The heatmaps show the proportion of high-confidence attention weights ($\alpha_{i,j} > \theta$) from each head that connects pairs of amino acids that are in contact with one another. In TapeBert (a), for example, we can see that 45% of attention in head 12-4 (the 12th layer's 4th head) maps to contacts. The bar plots show the maximum value from each layer. Note that the vertical striping in ProtAlbert (b) is likely due to cross-layer parameter sharing (see Appendix A.3).

respective datasets (note that none of the aforementioned annotations were used in model training). For the diagnostic classifier, we used the respective training splits for training and the validation splits for evaluation. See Appendix B.4 for additional details.

Experimental details We exclude attention to the `[SEP]` delimiter token, as it has been shown to be a "no-op" attention token (Clark et al., 2019), as well as attention to the `[CLS]` token, which is not explicitly used in language modeling. We only include results for attention heads where at least 100 high-confidence attention arcs are available for analysis. We set the attention threshold θ to 0.3 to select for high-confidence attention while retaining sufficient data for analysis. We truncate all protein sequences to a length of 512 to reduce memory requirements.[1]

We note that all of the above analyses are purely associative and do not attempt to establish a causal link between attention and model behavior (Vig et al., 2020; Grimsley et al., 2020), nor to *explain* model predictions (Jain & Wallace, 2019; Wiegreffe & Pinter, 2019).

4 What does attention understand about proteins?

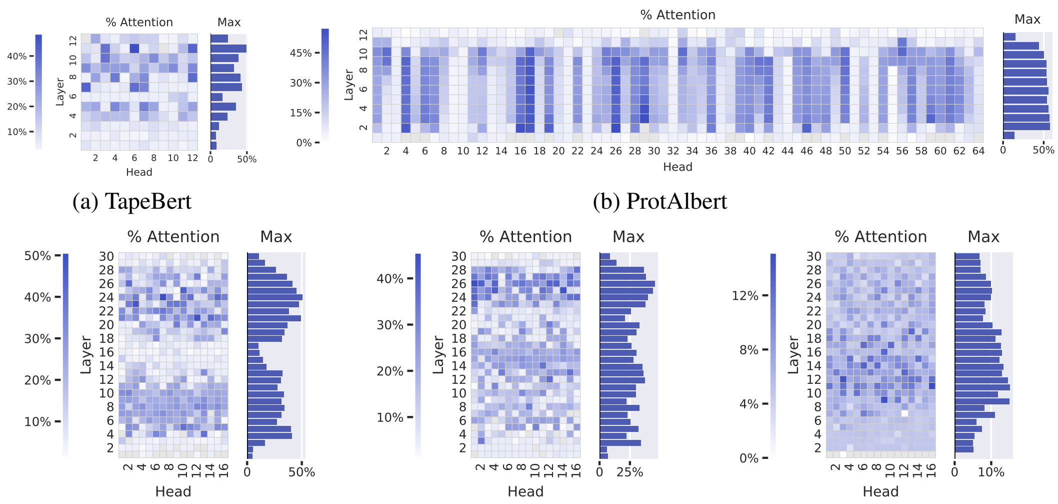
4.1 Protein Structure

Here we explore the relationship between attention and tertiary structure, as characterized by contact maps (see Section 2). Secondary structure results are included in Appendix C.1.

Attention aligns strongly with contact maps in the deepest layers. Figure 2 shows how attention aligns with contact maps across the heads of the five models evaluated[2], based on the metric defined in Equation 1. The most aligned heads are found in the deepest layers and focus up to 44.7% (TapeBert), 55.7% (ProtAlbert), 58.5% (ProtBert), 63.2% (ProtBert-BFD), and 44.5% (ProtXLNet) of attention on contacts, whereas the background frequency of contacts among all amino acid pairs in the dataset is 1.3%. Figure 1a shows an example of the induced attention from the top head in TapeBert. We note that the model with the single most aligned head—ProtBert-BFD—is the largest model (same size as ProteinBert) at 420M parameters (Appendix A.1) and it was also the only model pre-trained on the

[1]94% of sequences had length less than 512. Experiments performed on single 16GB Tesla V-100 GPU.

[2]Heads with fewer than 100 high-confidence attention weights across the dataset are grayed out.



(a) TapeBert (b) ProtAlbert

(c) ProtBert (d) ProtBert-BFD (e) ProtXLNet

Figure 3: Proportion of attention focused on binding sites across five pretrained models. The heatmaps show the proportion of high-confidence attention ($\alpha_{i,j} > \theta$) from each head that is directed to binding sites. In TapeBert (a), for example, we can see that 49% of attention in head 11-6 (the 11th layer's 6th head) is directed to binding sites. The bar plots show the maximum value from each layer.

largest dataset, BFD. It's possible that both factors helped the model learn more structurally-aligned attention patterns. Statistical significance tests and null models are reported in Appendix C.2.

Considering the models were trained on language modeling tasks without any spatial information, the presence of these structurally-aware attention heads is intriguing. One possible reason for this emergent behavior is that contacts are more likely to biochemically interact with one another, creating statistical dependencies between the amino acids in contact. By focusing attention on the contacts of a masked position, the language models may acquire valuable context for token prediction.

While there seems to be a strong correlation between the attention head output and classically-defined contacts, there are also differences. The models may have learned differing contextualized or nuanced formulations that describe amino acid interactions. These learned interactions could then be used for further discovery and investigation or repurposed for prediction tasks similar to how principles of coevolution enabled a powerful representation for structure prediction.

4.2 Binding Sites and Post-Translational Modifications

We also analyze how attention interacts with binding sites and post-translational modifications (PTMs), which both play a key role in protein function.

Attention targets binding sites throughout most layers of the models. Figure 3 shows the proportion of attention focused on binding sites (Eq. 1) across the heads of the 5 models studied. Attention to binding sites is most pronounced in the ProtAlbert model (Figure 3b), which has 22 heads that focus over 50% of attention on bindings sites, whereas the background frequency of binding sites in the dataset is 4.8%. The three BERT models (Figures 3a, 3c, and 3d) also attend strongly to binding sites, with attention heads focusing up to 48.2%, 50.7%, and 45.6% of attention on binding sites, respectively. Figure 1b visualizes the attention in one strongly-aligned head from the TapeBert model. Statistical significance tests and a comparison to a null model are provided in Appendix C.3.

ProtXLNet (Figure 3e) also targets binding sites, but not as strongly as the other models: the most aligned head focuses 15.1% of attention on binding sites, and the average head directs just 6.2% of attention to binding sites, compared to 13.2%, 19.8%, 16.0%, and 15.1% for the first four models in Figure 3. It's unclear whether this disparity is due to differences in architectures or pre-training objectives; for example, ProtXLNet uses a bidirectional auto-regressive pretraining method (see Appendix A.2), whereas the other 4 models all use masked language modeling objectives.

Figure 4: Each plot shows the percentage of attention focused on the given property, averaged over all heads within each layer. The plots, sorted by center of gravity (red dashed line), show that heads in deeper layers focus relatively more attention on binding sites and contacts, whereas attention toward specific secondary structures is more even across layers.

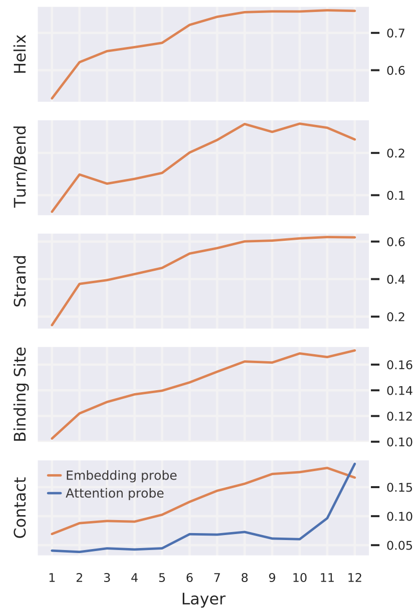


Figure 5: Performance of probing classifiers by layer, sorted by task order in Figure 4. The embedding probes (orange) quantify the knowledge of the given property that is encoded in each layer's output embeddings. The attention probe (blue), show the amount of information encoded in attention weights for the (pairwise) contact feature. Additional details are provided in Appendix B.3.

Why does attention target binding sites? In contrast to contact maps, which reveal relationships *within* proteins, binding sites describe how a protein interacts with other molecules. These external interactions ultimately define the high-level function of the protein, and thus binding sites remain conserved even when the sequence as a whole evolves (Kinjo & Nakamura, 2009). Further, structural motifs in binding sites are mainly restricted to specific families or superfamilies of proteins (Kinjo & Nakamura, 2009), and binding sites can reveal evolutionary relationships among proteins (Lee et al., 2017). Thus binding sites may provide the model with a high-level characterization of the protein that is robust to individual sequence variation. By attending to these regions, the model can leverage this higher-level context when predicting masked tokens throughout the sequence.

Attention targets PTMs in a small number of heads. A small number of heads in each model concentrate their attention very strongly on amino acids associated with post-translational modifications (PTMs). For example, Head 11-6 in TapeBert focused 64% of attention on PTM positions, though these occur at only 0.8% of sequence positions in the dataset.[3] Similar to our discussion on binding sites, PTMs are critical to protein function (Rubin & Rosen, 1975) and thereby are likely to exhibit behavior that is conserved across the sequence space. See Appendix C.4 for full results.

4.3 Cross-Layer Analysis

We analyze how attention captures properties of varying complexity across different layers of TapeBert, and compare this to a probing analysis of embeddings and attention weights (see Section 3).

Attention targets higher-level properties in deeper layers. As shown in Figure 4, deeper layers focus relatively more attention on binding sites and contacts (high-level concept), whereas secondary structure (low- to mid-level concept) is targeted more evenly across layers. The probing analysis of attention (Figure 5, blue) similarly shows that knowledge of contact maps (a pairwise feature)

[3]This head also targets binding sites (Fig. 3a) but at a percentage of 49%.